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Lecture #7: DeepFake Generation and Detection

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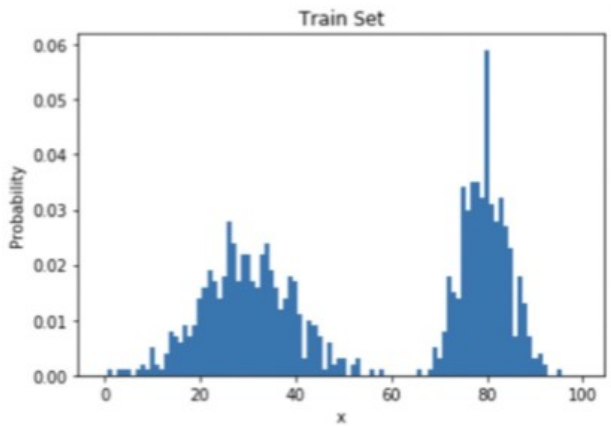
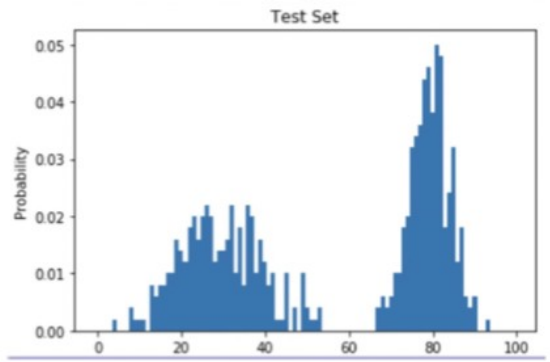
Recall: Data Synthesis

- Lets say we have some training set of data following some distribution $p_{data}(x)$:





We generate new data by *sampling* from the learned distribution

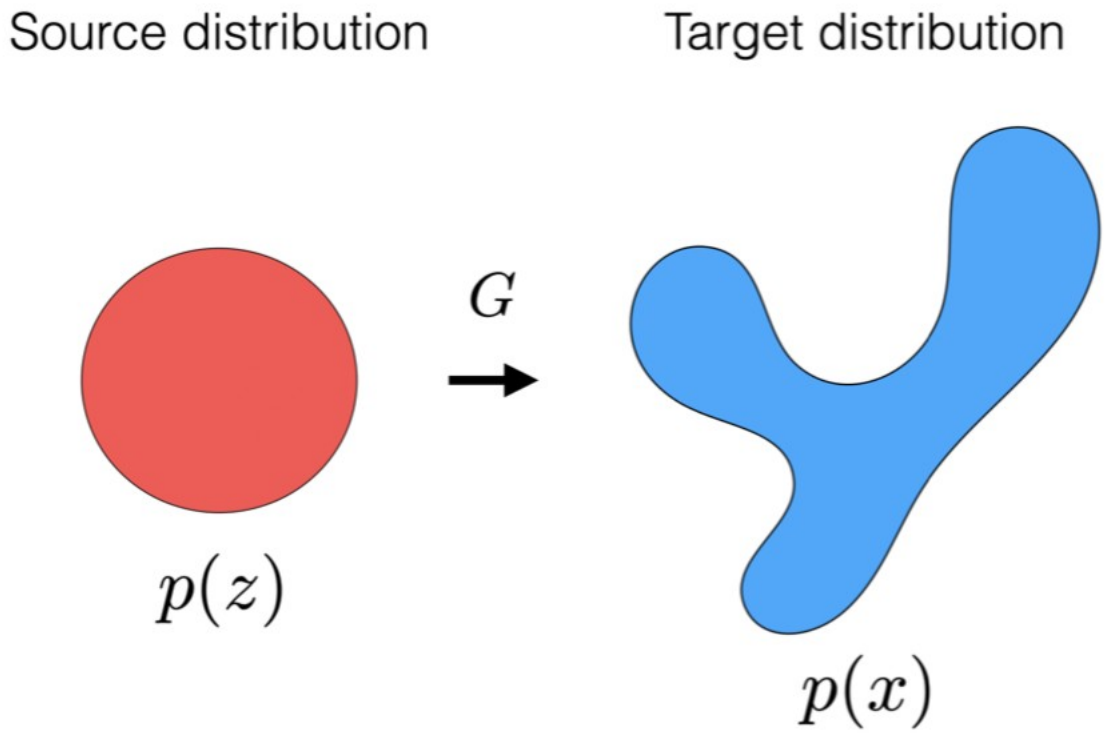
The goal of generative machine learning models is to learn this distribution to the best of their ability

Methods for Learning Data Distribution

- (1) We want to train models that maximize the expected log likelihood of $p_{data}(x)$. If I sample from the distribution and get a
- ● high likelihood--- likely the sample came from the training distribution
- ●low likelihood--- the sample probably didn't come from the training distribution

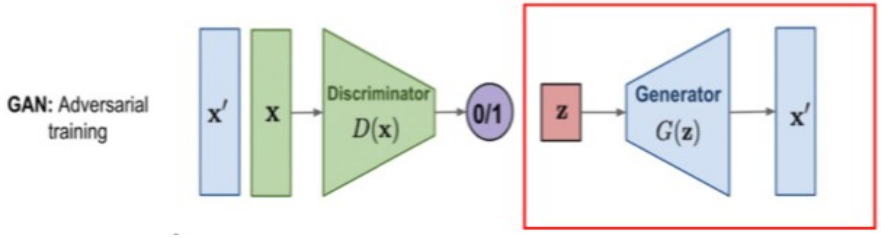
- (2) We want to minimize some divergence metrics between the training data distribution, and the distribution that the model learns

Sampling from Noise



GANs

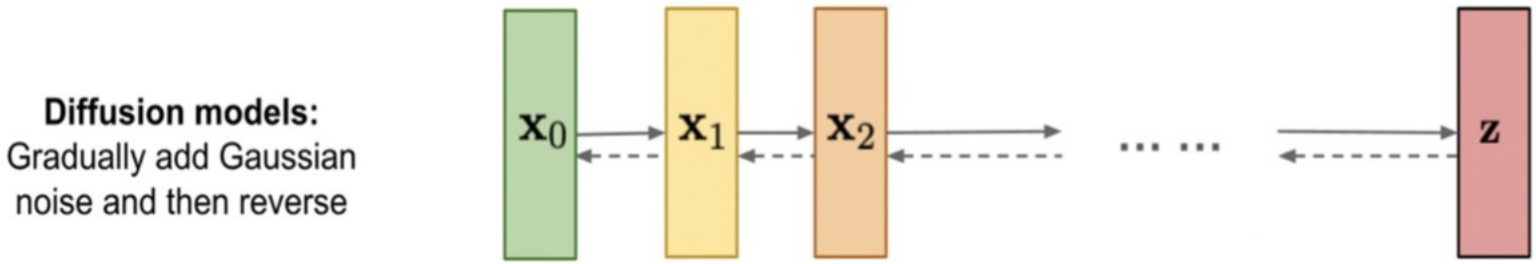
- Convert latent noise vector to target distribution in one step



- Lots of issues with training:
  - ○Vanishing gradients: if discriminator is too good, gradients go to zero
  - ○Mode collapse: if the generator learns to generate an exceptionally plausible output, it will just continue generating it. Then the discriminator will learn to always reject it, and then the generator will produce the same outputs, which the discriminator will then reject… bad loop!

Diffusion

- Idea: Estimating and analyzing small step sizes is more tractable/easier than a single step from random noise to the learned distribution
- Convert a well-known and simple base distribution (like a Gaussian) to the target (data) distribution iteratively, with small step sizes, via a Markov chain:



- Markov chain: outlines the probability associated with a sequence of events occurring based on the state in the previous event.

Forward Process

- Noise added can be parameterized by:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \quad q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}) \quad \{\beta_t \in (0,1)\}_{t=1}^{T}$$

Vary the parameters of the Gaussian according to a *noise schedule*

- You can prove with some math that as T approaches infinity, you eventually end up with an Isotropic Gaussian (i.e. pure random noise)
- ●Note: forward process is fixed

Reparameterization trick

- Do you have to add noise iteratively to get to some timestep !? Nope!
- Reverse process can be written in one step:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0,\ (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$\alpha_t = 1 - \beta_t$$
$$\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$$

This will be useful during training!

Implementing Forward Process

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$\alpha_t = 1 - \beta_t$$
$$\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$$

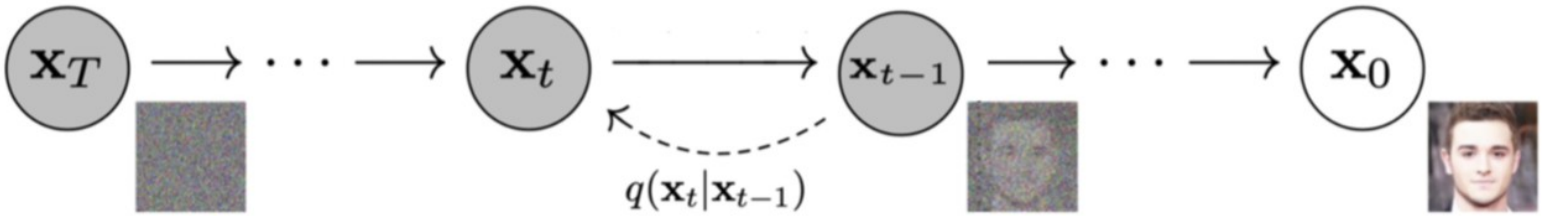
1. Sample an image from the dataset:
2. Sample noise $\epsilon \sim N(0, \mathbf{I})$ (from a **standard** normal distribution)
3. Scale the image by $\sqrt{\overline{\alpha_t}}$: $\sqrt{\overline{\alpha_t}}\, x_0$

where $\alpha_t = 1 - \beta_t$
$\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$

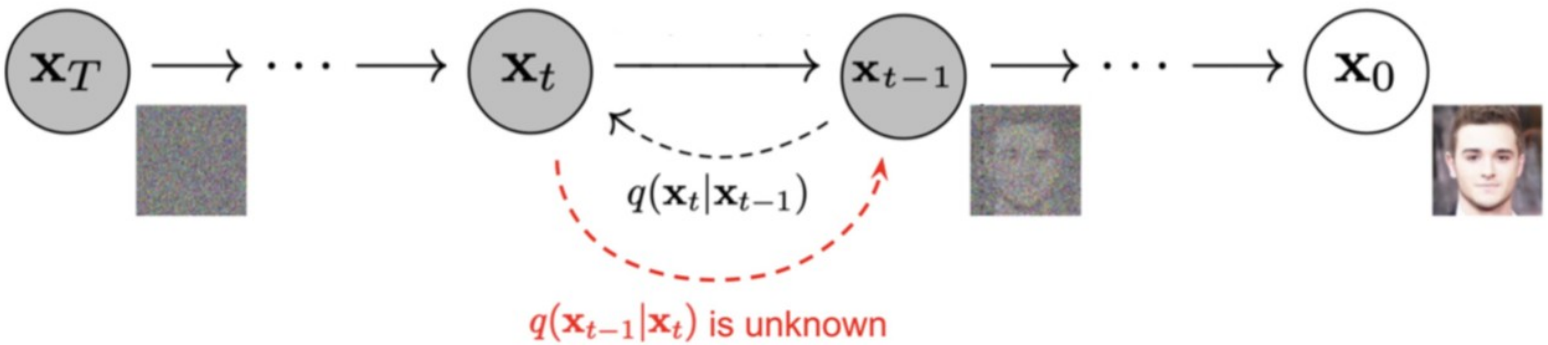
4. Add $\sqrt{1-\overline{\alpha_t}}\, \epsilon$: $\sqrt{\overline{\alpha_t}}\, x_0 + \sqrt{1-\overline{\alpha_t}}\, \epsilon$ $\longrightarrow$

x_t

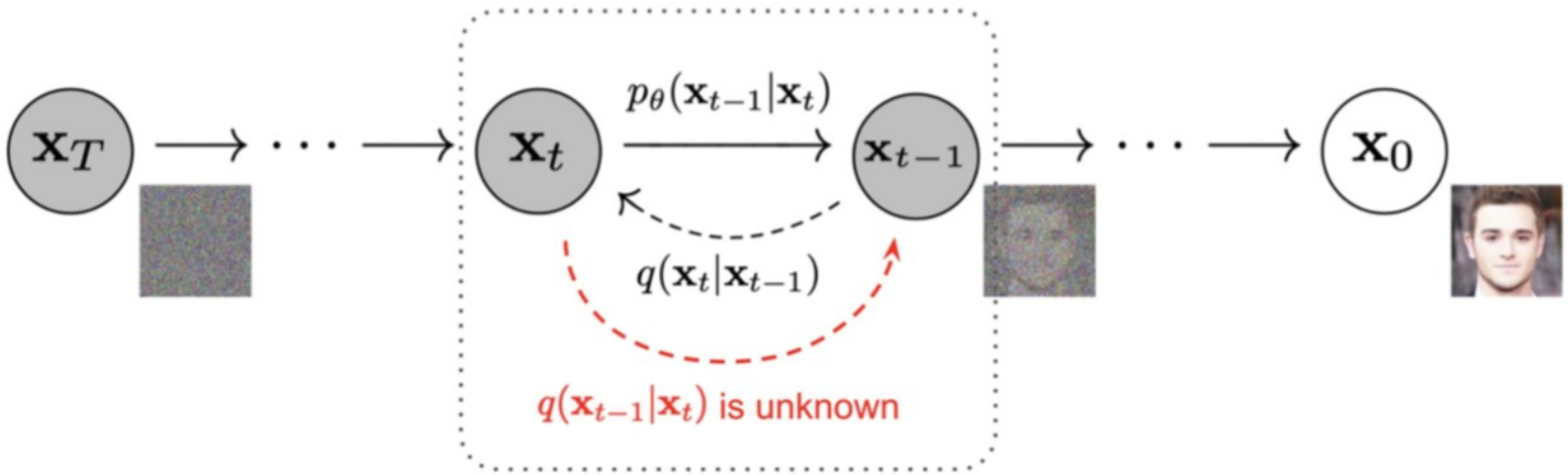
Reverse Process



Reverse Process



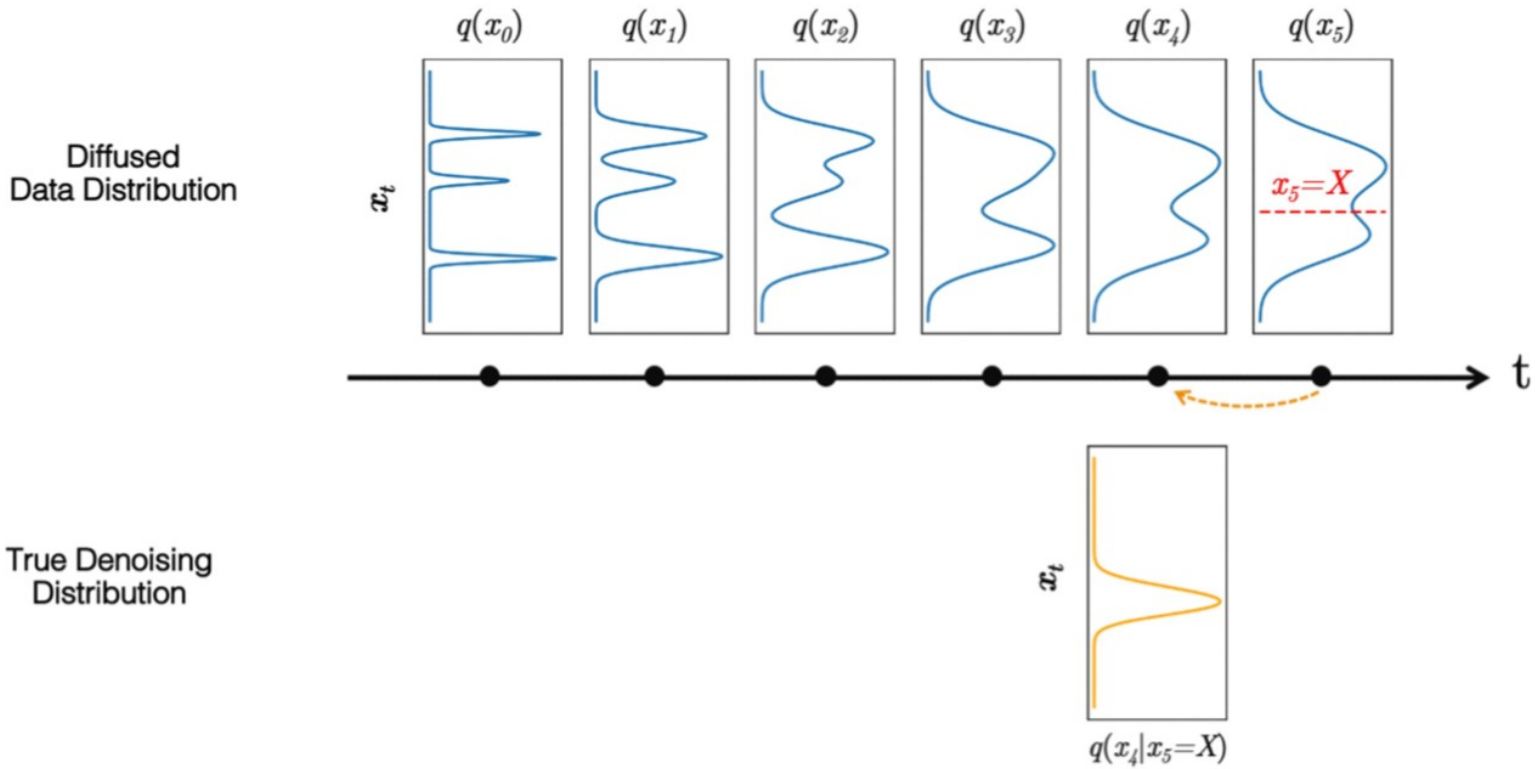
Reverse Process

- 

- The goal of a diffusion model is to learn the reverse denoising process to iteratively undo the forward process
- In this way, the reverse process appears as if it is generating new data from random noise!

Diffused data distribution

Diffused data distribution



What should the distribution look like?

Turns out that for small enough forward steps, i.e. $\{\beta_t \in (0,1)\}_{t=1}^{T}$

the reverse process step $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ can be estimated as a Gaussian distribution too

Therefore, we can parametrize the *learned* reverse process as

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$

In practice, Σ is just the identify matrix, so we only need to learn the mean of the distribution

Preliminary objective

When we write out the loss function, we get something that looks like this:

$$
\begin{aligned}
L_{\mathrm{VLB}} &= L_T + L_{T-1} + \cdots + L_0 \\
\text{where } L_T &= D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T)) \\
L_t &= D_{\mathrm{KL}}(q(\mathbf{x}_t|\mathbf{x}_{t+1}, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1})) \text{ for } 1 \leq t \leq T-1 \\
L_0 &= -\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)
\end{aligned}
$$

Middle Loss Term - Intuition

- KL Divergence: measures distance between two distributions
- If high, very dissimilar distributions
- If low, very similar distributions
- Goal: drive this very low

$$L_t = D_{\mathrm{KL}}(q(\mathbf{x}_t|\mathbf{x}_{t+1}, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1})) \text{ for } 1 \leq t \leq T-1$$

Final Loss

- Recall, our goal was to learn the following /! (network that parameterizes the mean of the data distribution):

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$

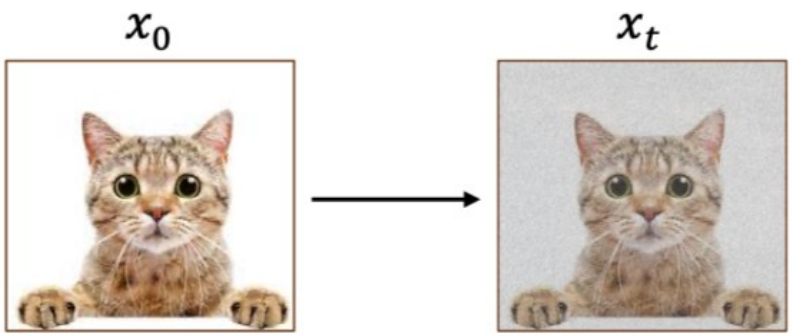
So we minimize:

$$\text{MSE}(\mu_\theta(x_t, t), x_{t-1})$$

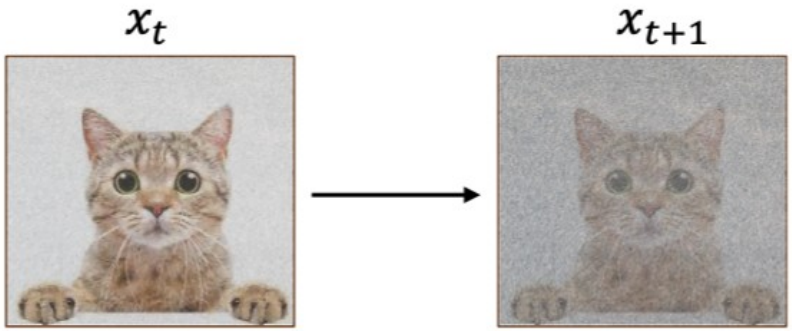
How do we do this in practice?

Step 1: Sample image from the dataset, generate noisy image using forward process

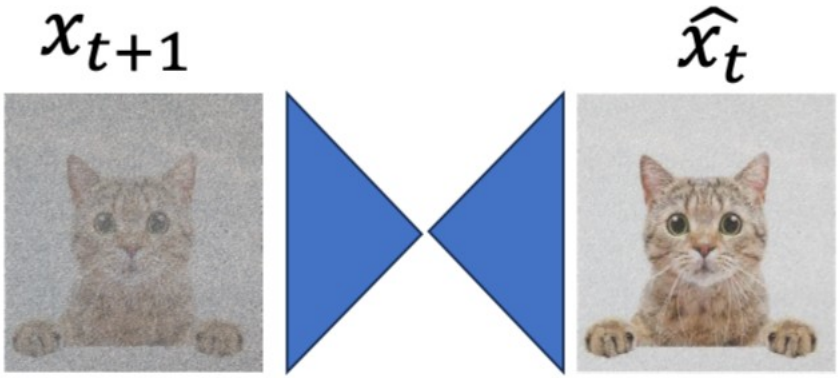
$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0,\ (1-\bar{\alpha}_t)\mathbf{I}\right)$$



Step 2: Given noisy image, generate slightly noisier image

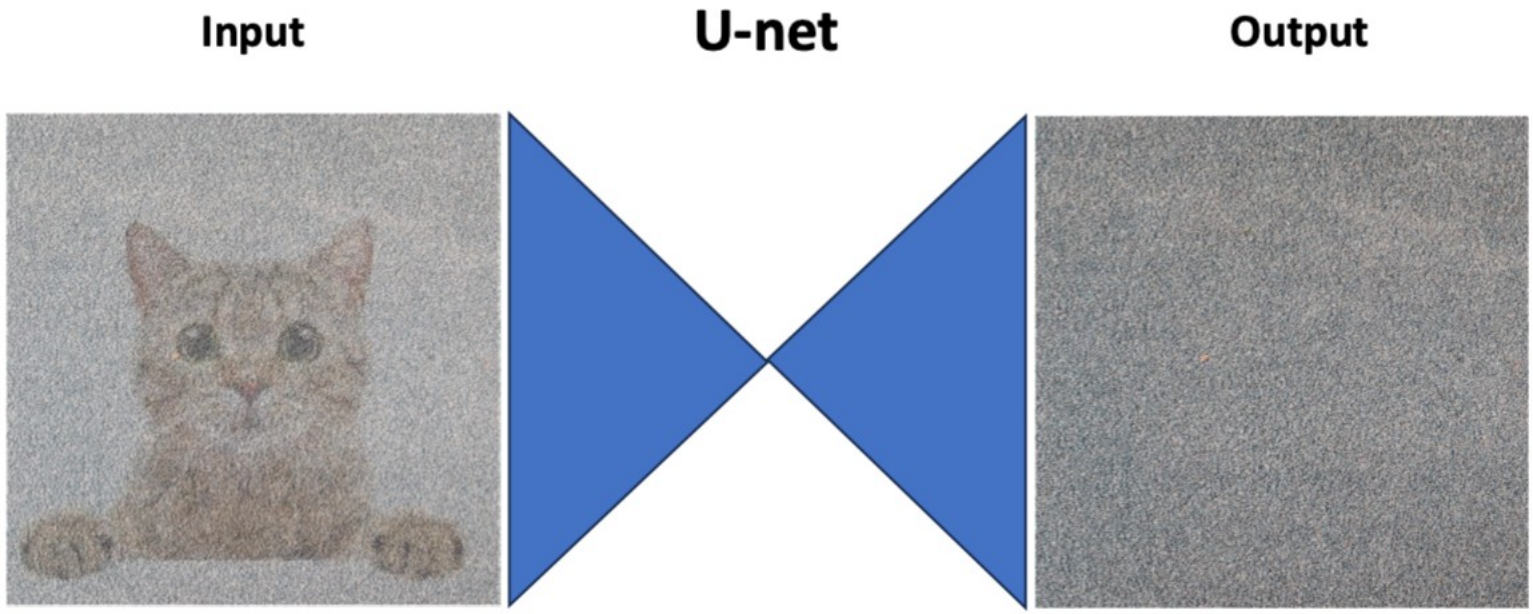


How do we do this in practice?



Loss: MSE$(x_t, \widehat{x_t})$

Neural Network that predicts noise



Training

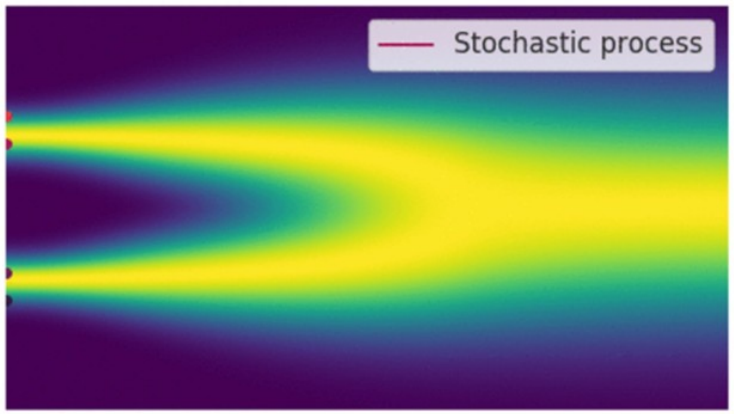
Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on

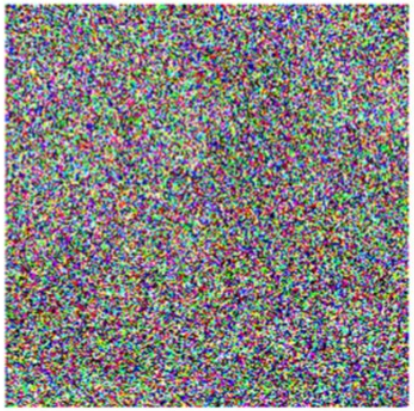
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$

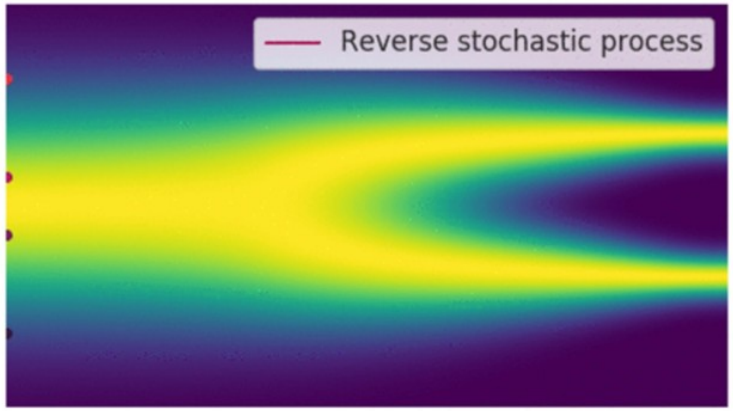
6: **until** converged

Process



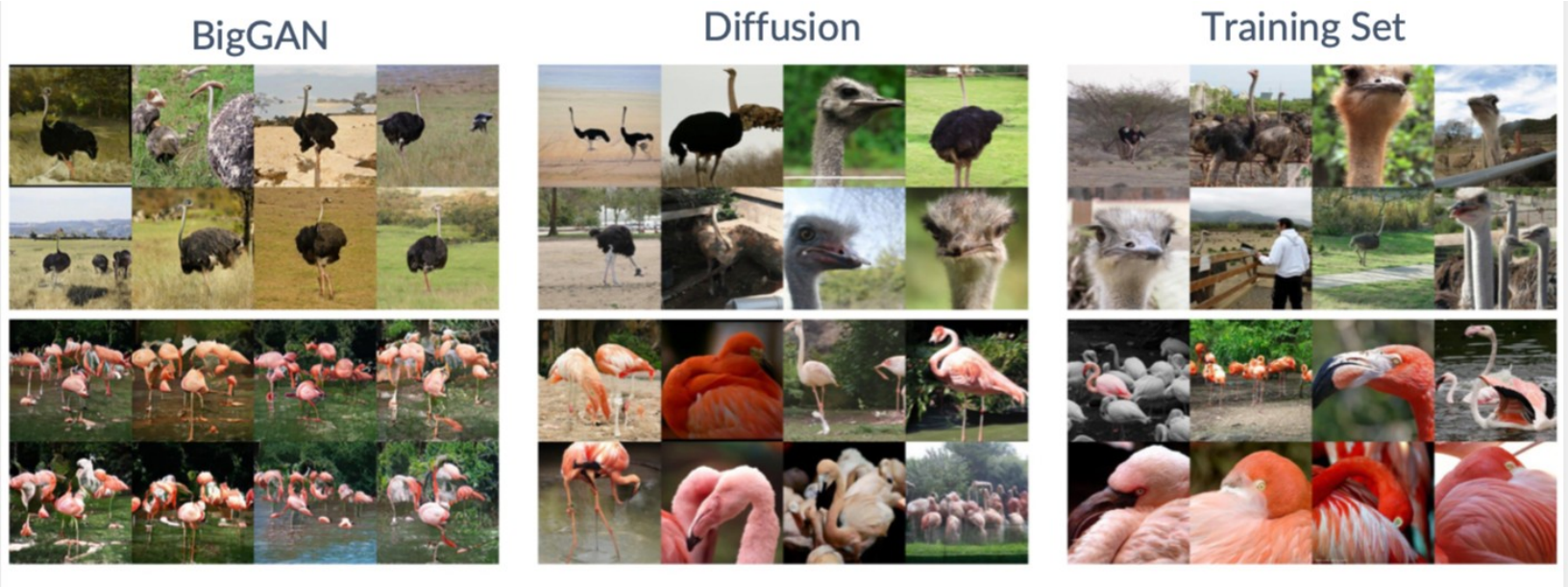
Forward process: converting the image distribution to pure noise





Reverse process: sampling from the image distribution, starting with pure noise

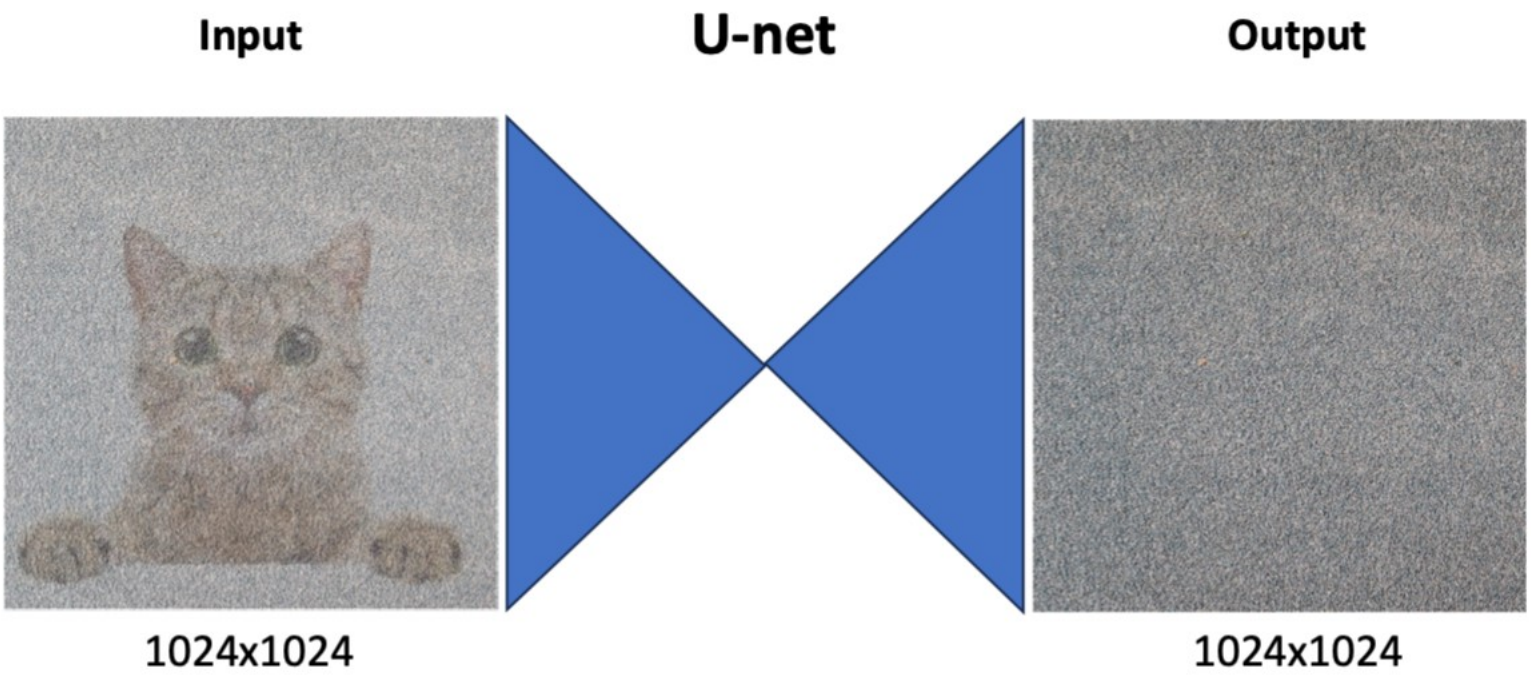
Diffusion Models Beats GANs



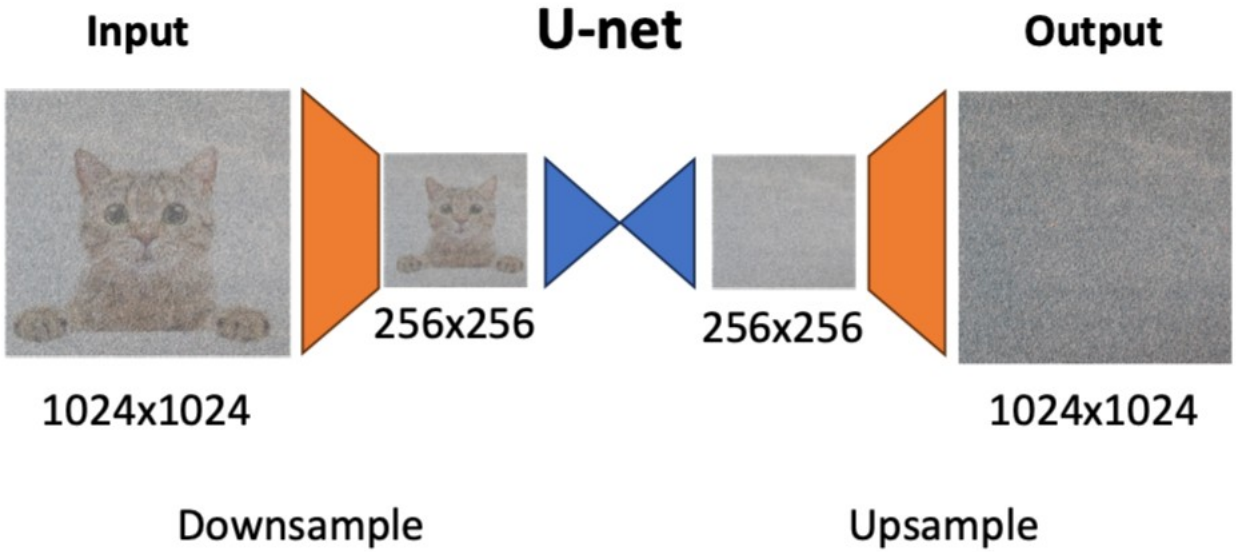
Diffusion Models Beats GANs

U-net Problem

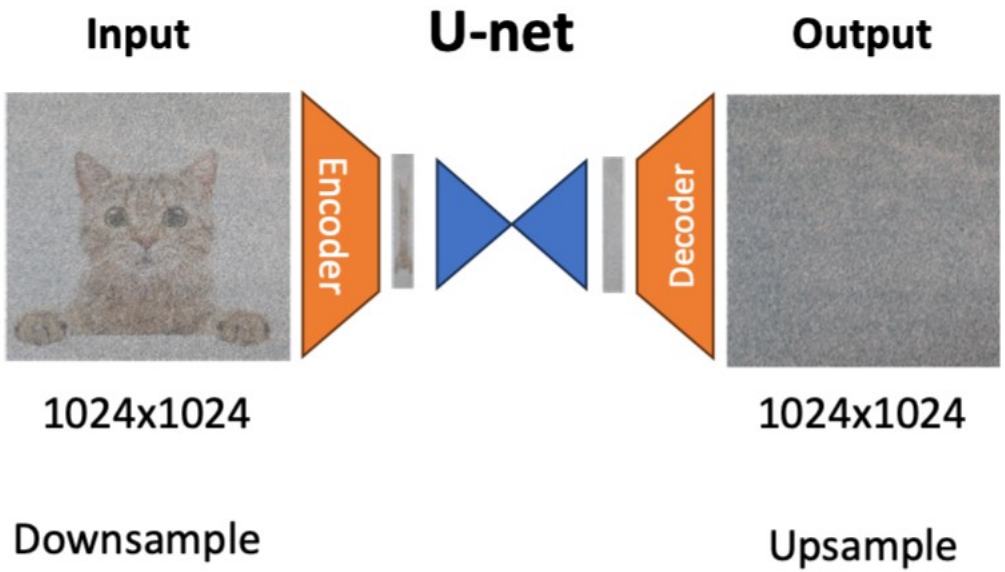
- Problem: operating in the input space is very computationally expensive!



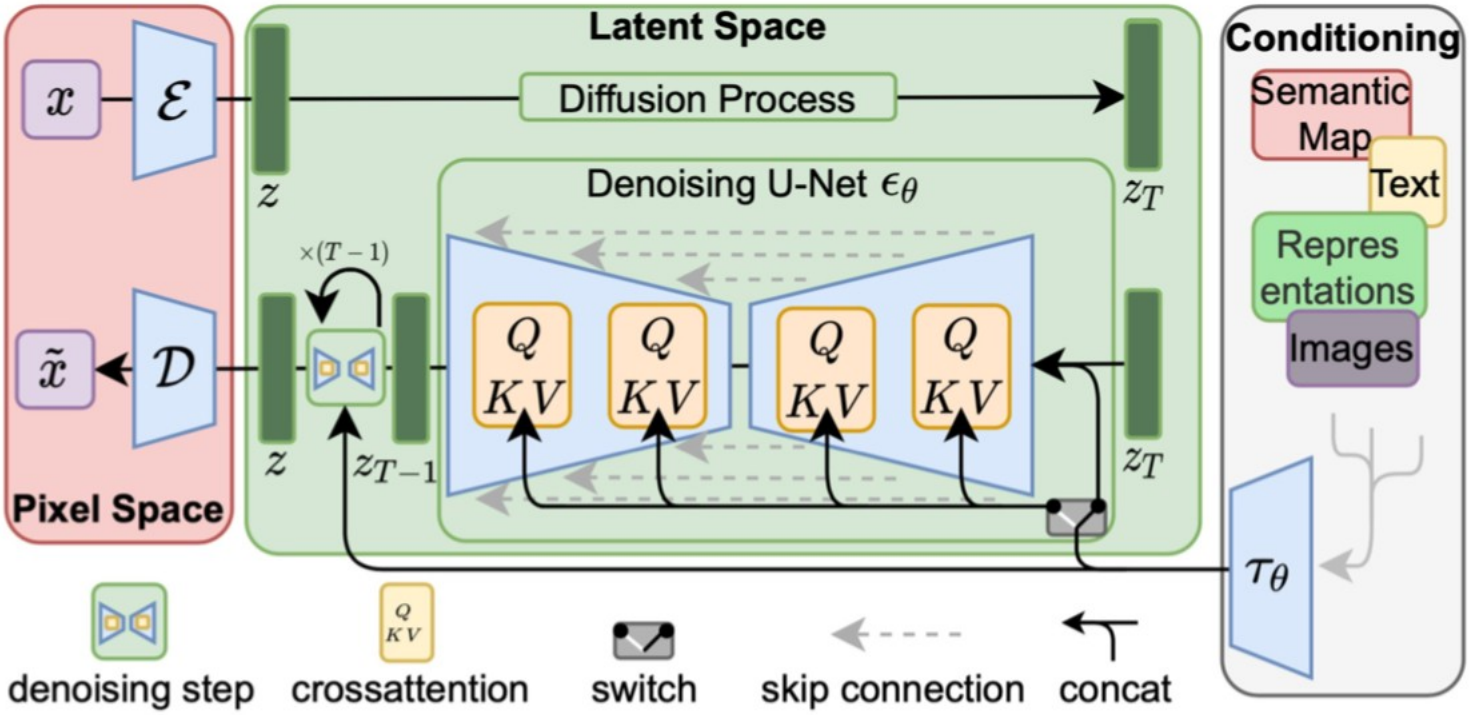
Option #1: Generate Low-Resolution + Upsample



Option #2: Generate in Latent Space



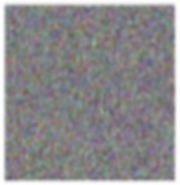
Stable Diffusion



Guided/Conditioned Diffusion

- Lets say we train a diffusion model on images of cats and dogs:

If we start from random noise, and generate a new image, what will the model generate?



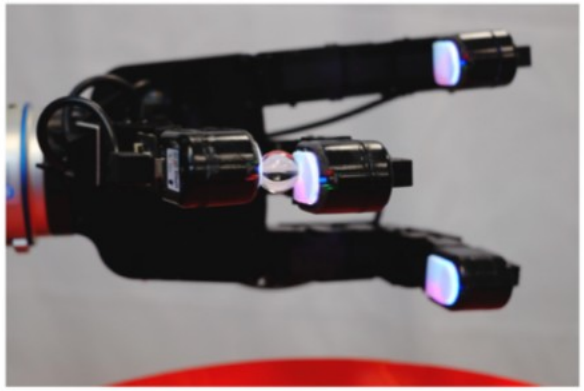
→ ?

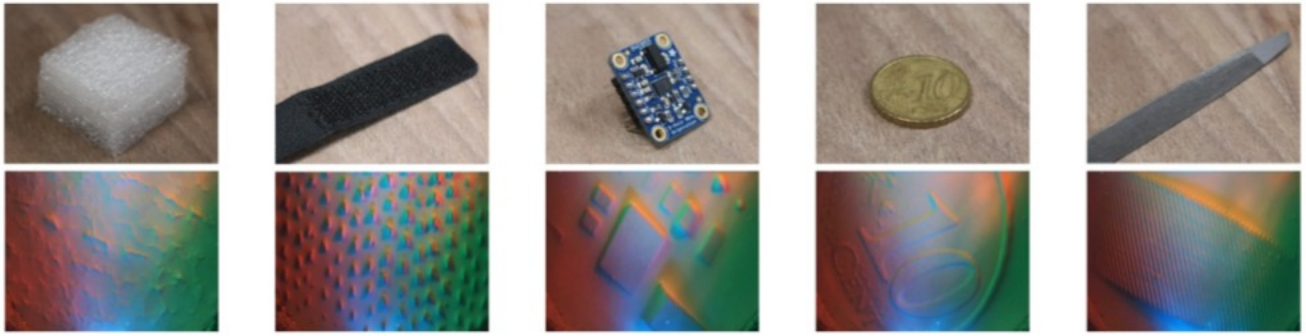
Leveraging Diffusion Models for Visual-Tactile Cross Generation

- 1.Background
- 2. Previous Methods
- 3. Learning to Read Braille (Vision-to-Touch)
- 4. Generating Visual Scenes from Touch (Touch-to-Vision)

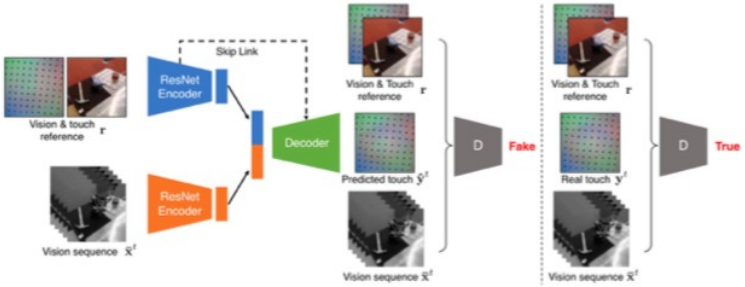
1 Background: Tactile Sensor & Touch Images

Figure 2: Exploded view of a single DIGIT sensor. A) elastomer, B) acrylic window, C) snap-fit holder, D) lighting PCB, E) plastic housing, F) camera PCB, G) back housing.

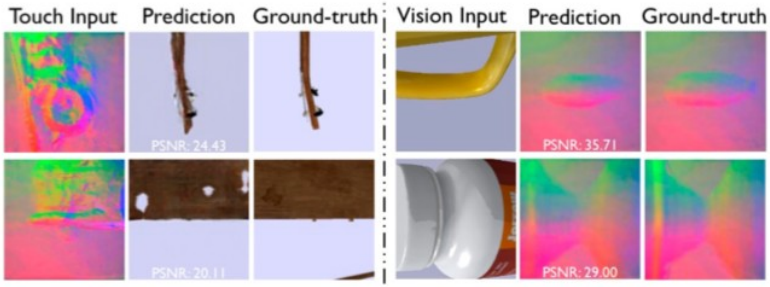




2 Previous Methods

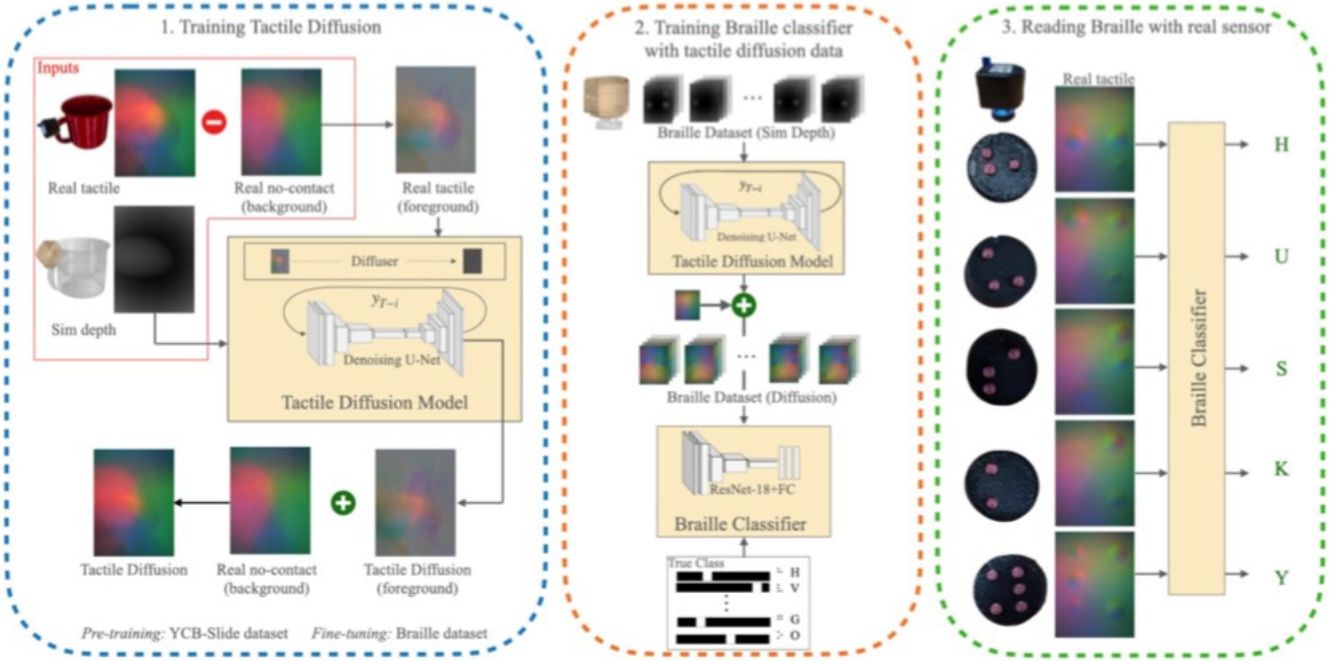


VisGel [1]: GAN-based exocentric generation



ObjectFolder [2]: GAN-based egocentric generation

3 Learning to Read Braille (Vision-to-Touch)



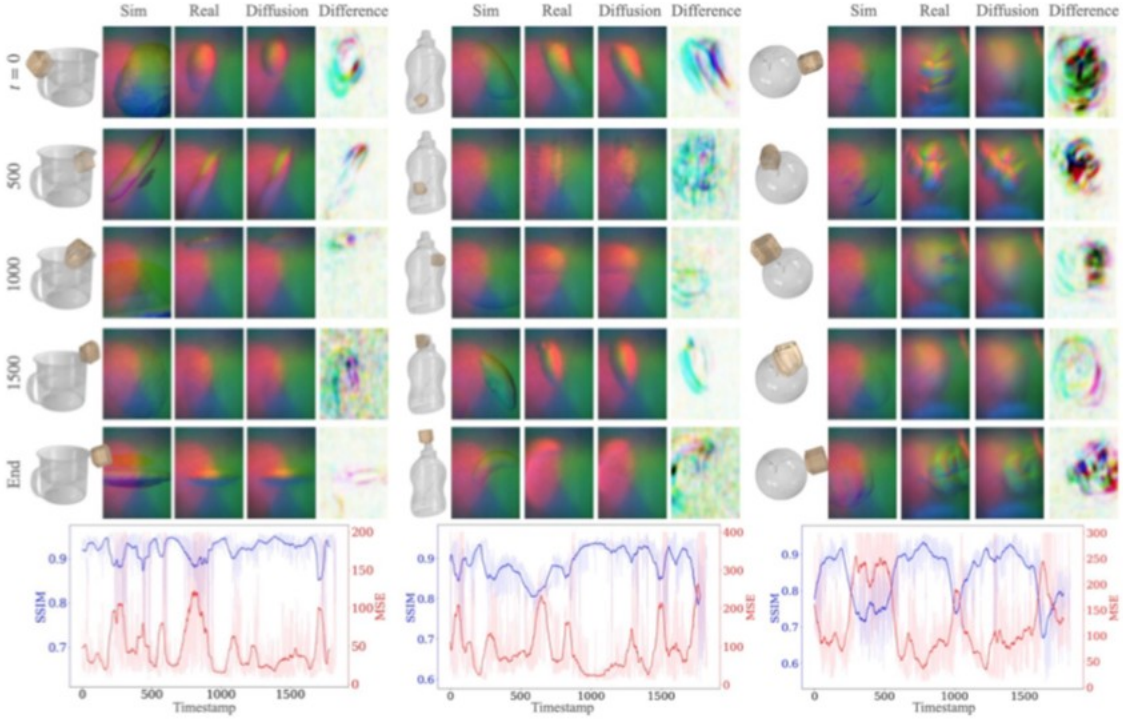
3 Learning to Read Braille (Vision-to-Touch)

Data: Simulated local depth map & Real tactile images collected on YCB dataset

3.1 Training Tactile Diffusion

- Data:
  - Simulated local depth map & Real tactile images collected on YCB dataset
- Diffusion decoder:
  - Conditional U-Net backbone that takes depth map as input and renders colorful tactile images
- Evaluation:
  - SSIM (structural similarity) & MSE (mean squared error)

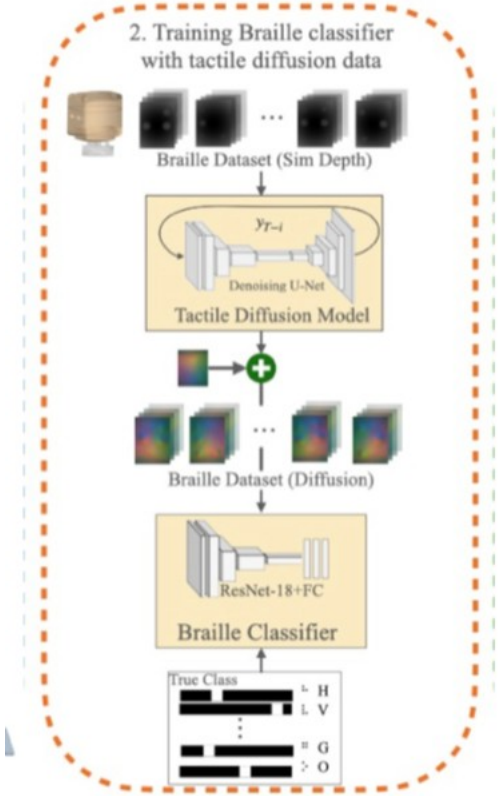
3.1 Training Tactile Diffusion



Simulation, real, tactile diffusion results
(SSIM is generally above 0.80)

3.2 Training Braille Classifier in Simulator

- Sim2Real Transfer:
  - Train a classifier to detect real-world braille letters with DIGIT sensor
- Comparison:
  - Compare results from Sim / cGAN / Diffusion / Real data.

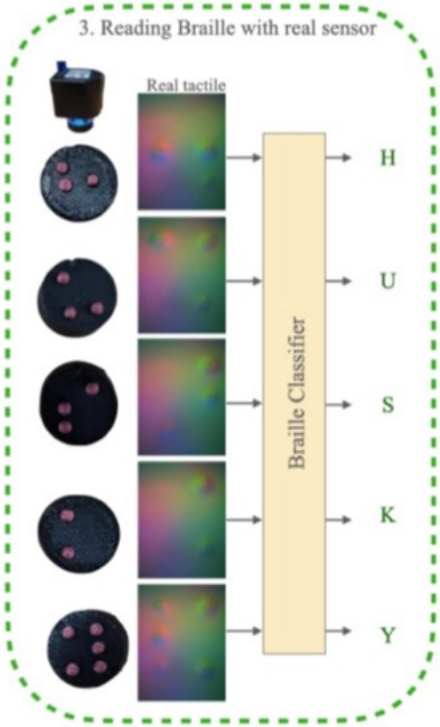


3.3 Reading Braille with Real-World Sensor

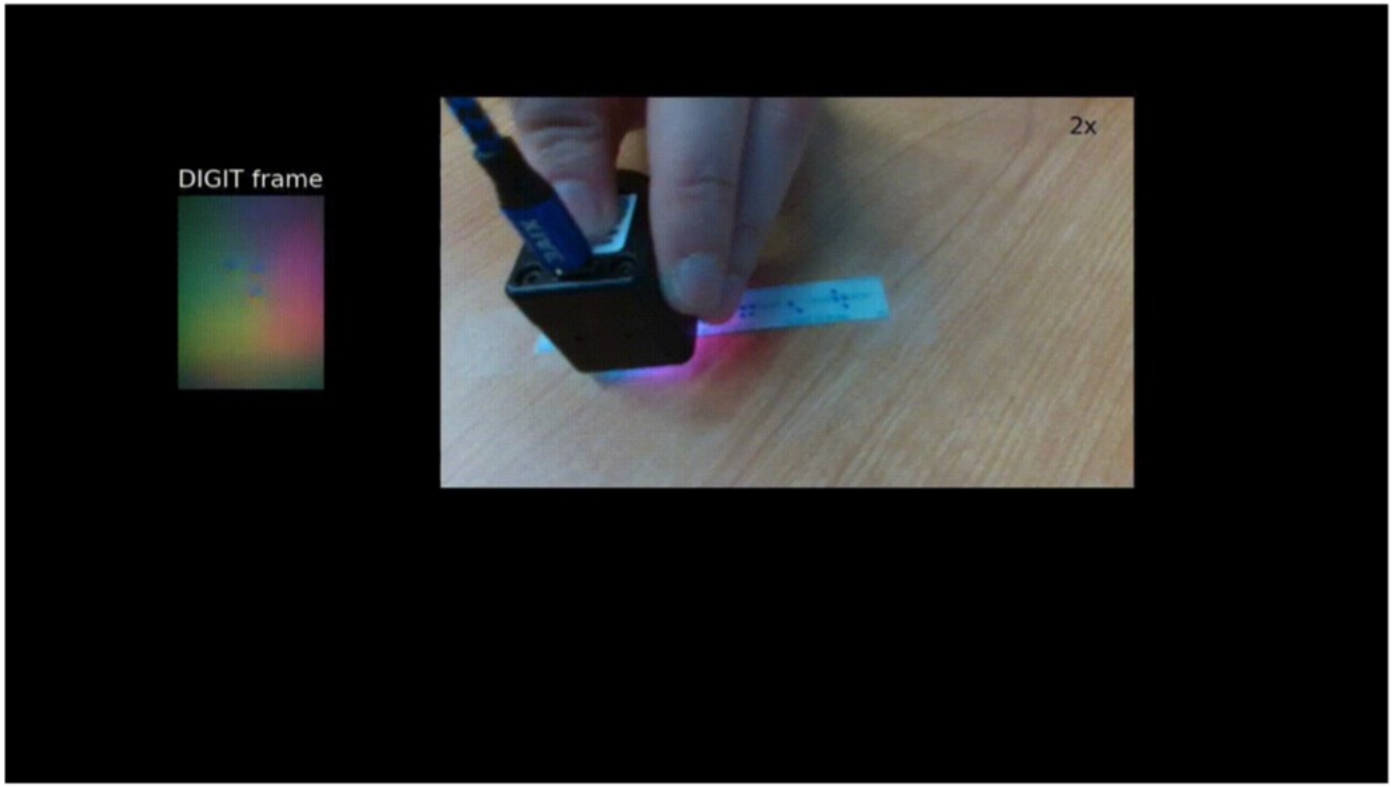
TABLE I: Metrics on braille classification task.

Training data source	% real data fine-tuning	Accuracy %	Precision	Recall
Sim	-	30.23	0.34	0.30
	20	64.99	0.71	0.65
	80	73.11	0.80	0.73
	100	73.95	**0.81**	0.74
Sim + data aug.	-	43.48	0.61	0.43
	100	73.23	0.76	0.73
cGAN	-	31.18	0.40	0.31
Tactile diffusion	-	**75.74**	0.79	**0.76**
Real	-	100.0	1.00	1.00

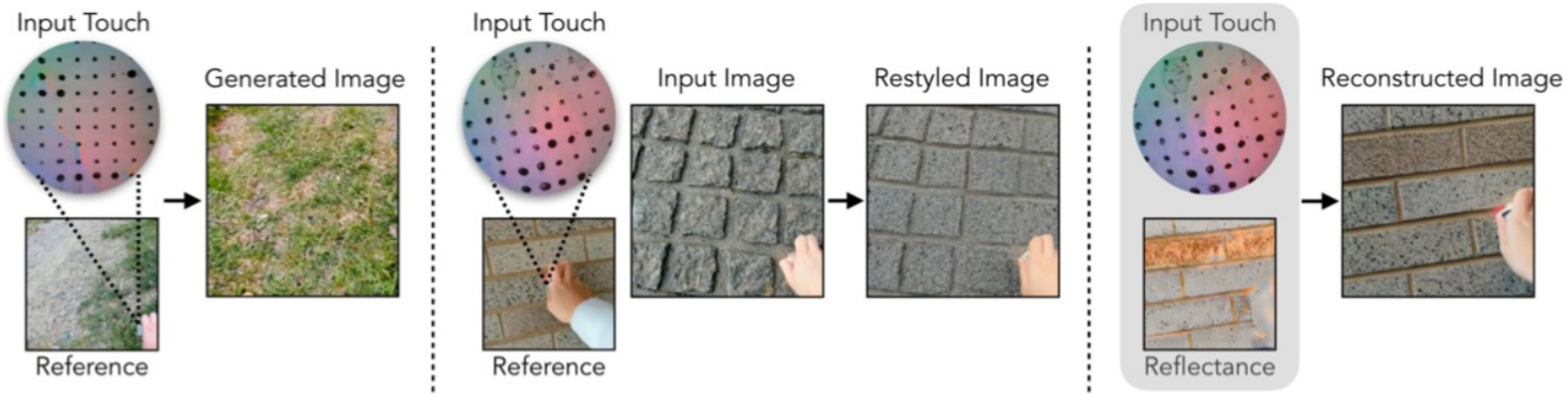
Training cGAN on 100% real, tactile diffusion on YCB-Slide + 20% real



3.3 Reading Braille with Real-World Sensor



4 Generating Visual Scenes from Touch (Touch-to-Vision)

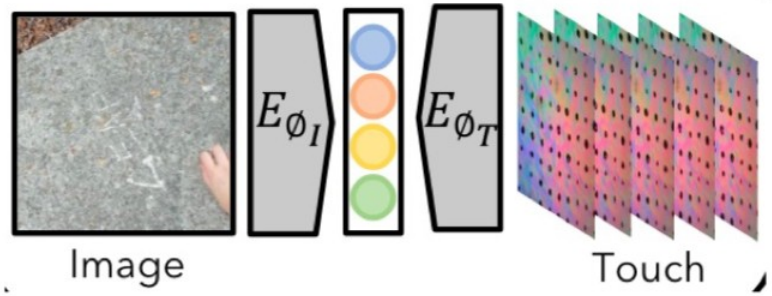


(a) Touch-to-Image Generation (b) Tactile-driven Image Stylization (c) Tactile-driven Shading Estimation

4.1 Contrastive Visuo-tactile Pretraining (CVTP)

Given N visual-tactile image pairs, sample K from them and perform contrastive learning (mapping them into the uniform hidden space) using InfoNCE Loss:

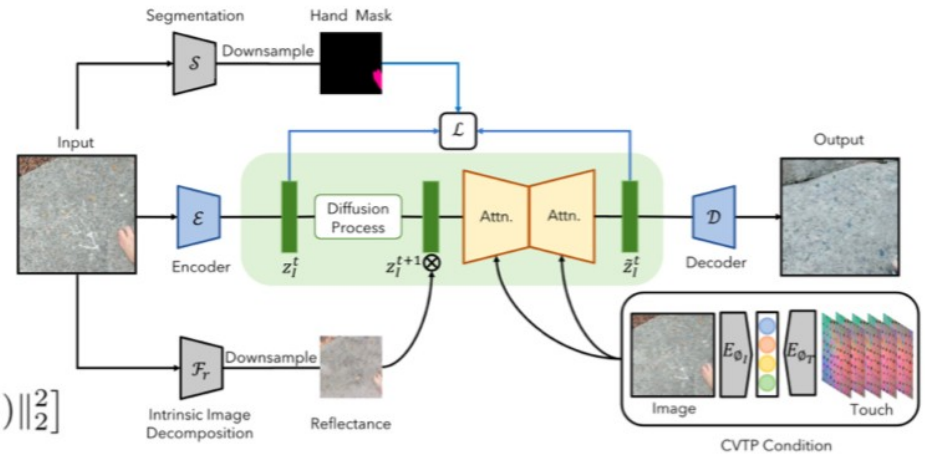
$$\mathcal{L}_i^{V_I,V_T} = -\log \frac{\exp(E_{\phi_I}(v_I^i) \cdot E_{\phi_T}(v_T^i)/\tau)}{\sum_{j=1}^{K} \exp(E_{\phi_I}(v_I^i) \cdot E_{\phi_T}(v_T^j)/\tau)} \quad (1)$$



4.2 Touch-conditioned Image Generation

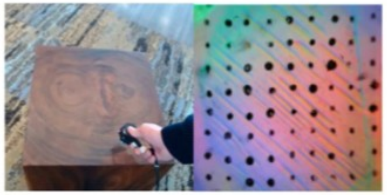
- Touch signal is represented by multi-frames from tactile sensor
- The diffusion process is conditioned on the touch signal. The loss function is:

$$L(\theta, \phi) = \mathbb{E}_{\mathbf{z}_I, \mathbf{c}, \epsilon, t} \left[\| \epsilon_t - \epsilon_\theta(\mathbf{z}_I^t, t, E_{\phi_T}(\mathbf{v}_T)) \|_2^2 \right]$$

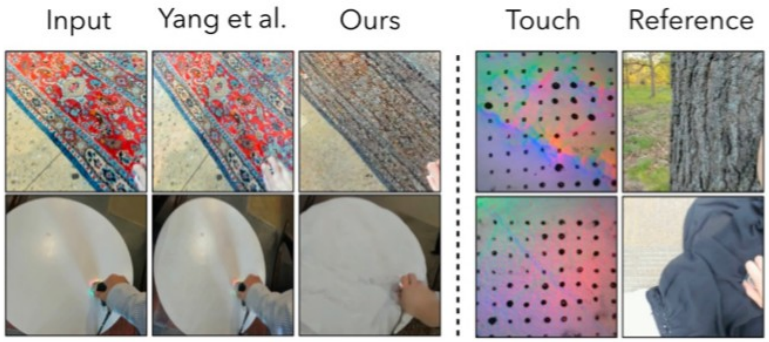


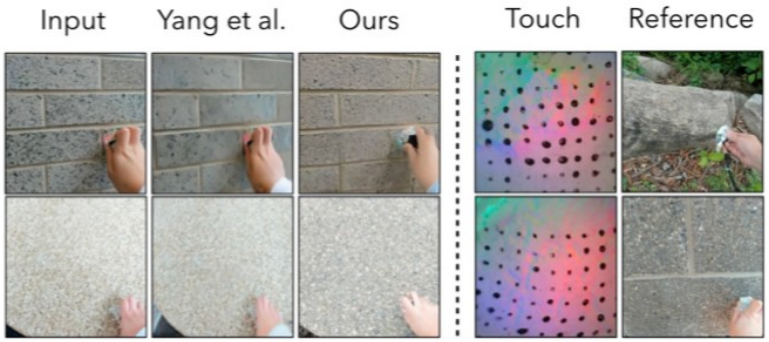
4.3 Qualitative Results: Tactile-driven Image Stylization

Touch and Go: An indoor-outdoor dataset with humans holding DIGIT sensor to touch objects



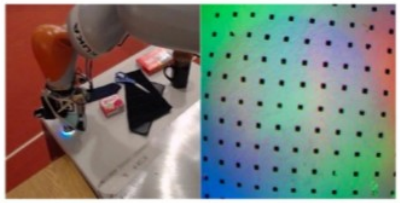
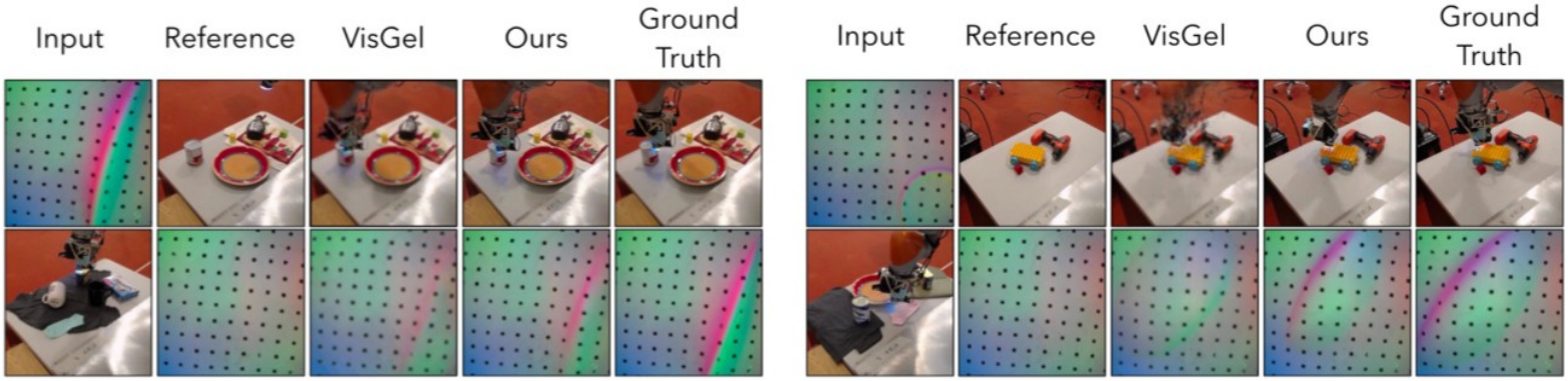
Touch and Go [64]





4.4 Qualitative Results: Visual-Tactile Cross Generation

VisGel: A dataset that collects paired touch videos and third-view robot arms.



VisGel [38]